

Social Media Impact Study

Teens, students, mental health, addiction, academic performance & suicide trend correlations.

3

datasets
analyzed

EDA

correlations, heatmaps &
boxplots

ML

logistic
regression model

Core question

How does social media usage relate to mental health, suicide trends, and academic outcomes?

MENTAL HEALTH**ACADEMICS****ADDICTION**

From raw data to insight.

The project combines national trend data, teen phone-addiction data, and student social-media-addiction data.

01

Collect

Three separate datasets on suicide trends, teens, and students.

02

Explore

EDA, correlation matrices, distributions, boxplots, and comparisons.

03

Model

Logistic regression predicts whether social media affects academics.

The analysis connects three perspectives.

Each dataset answers a different part of the same social-media impact question.

National trend data

Social media growth vs suicide-rate change since 2010.

Teen phone addiction

Usage hours, addiction level, sleep, anxiety, depression, and academics.

Student addiction data

Daily use, addiction score, mental health, sleep, conflicts, and academic impact.

mental health

anxiety, depression, wellbeing

academics

performance & self-reported impact

behavior

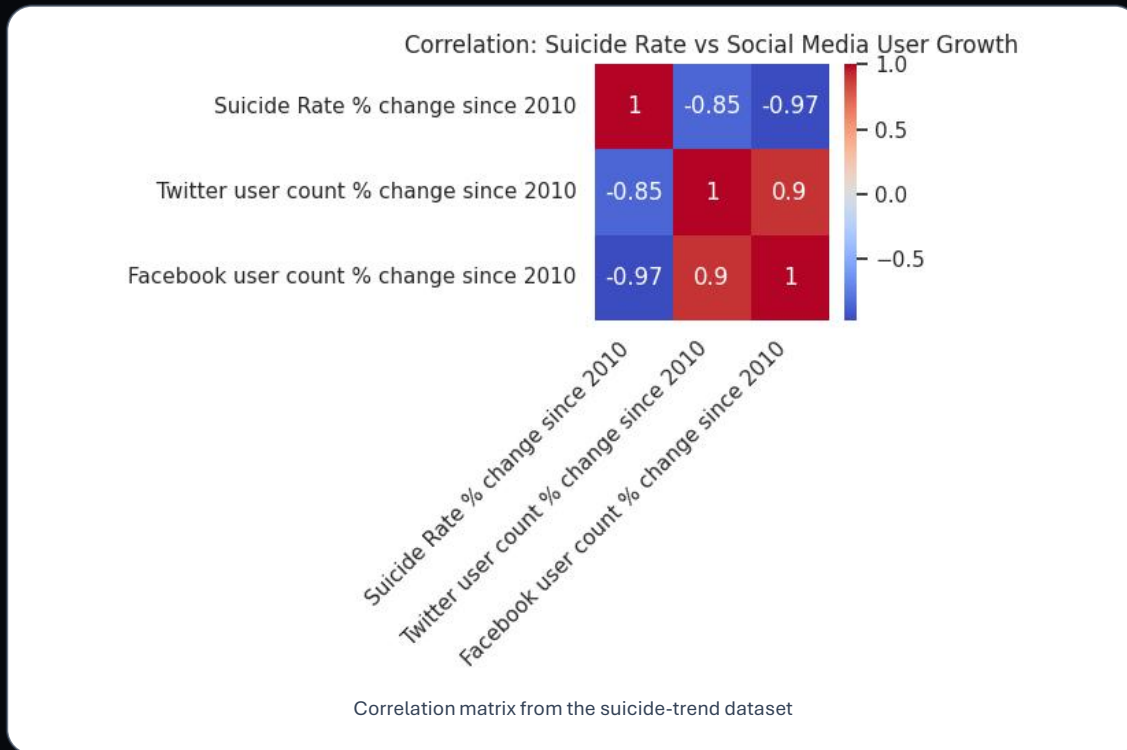
usage hours, addiction, sleep

trend

national suicide-rate correlation

National trend correlations were inverse.

Higher social-media user growth was associated with lower suicide-rate change in this dataset.



-0.97

Facebook growth vs suicide-rate change

-0.85

Twitter growth vs suicide-rate change

Important interpretation: correlation does not prove causation.
This finding only describes the pattern visible in the dataset.

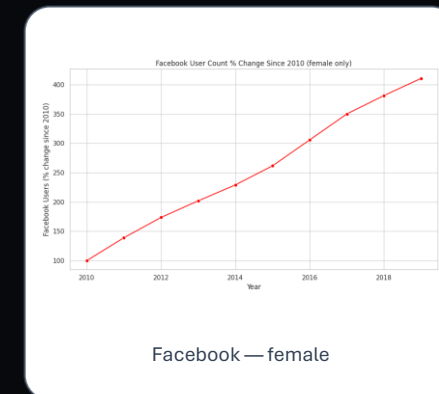
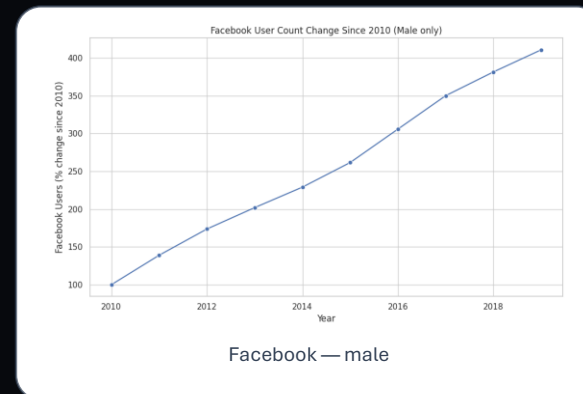
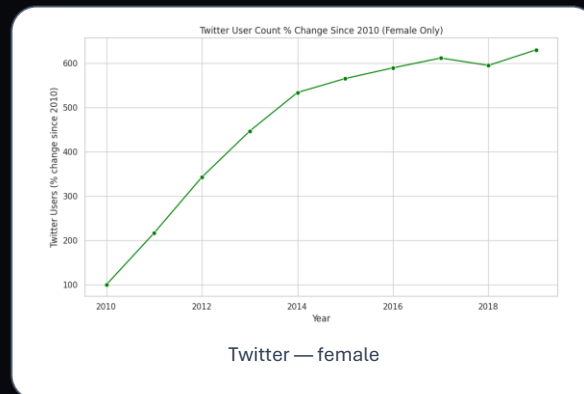
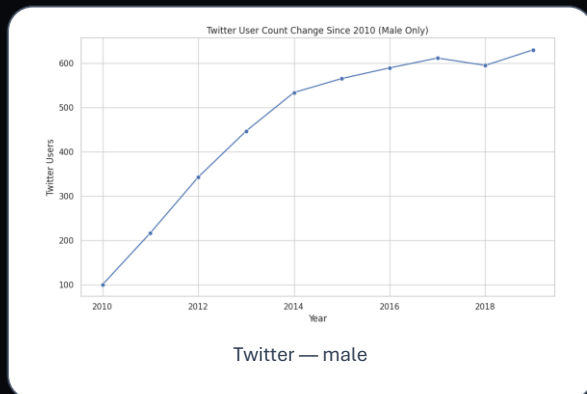
Why this matters

The national trend shows a clear pattern, but it does not explain what happens at the individual level.

Next, we examine teen and student data to explore social media use, addiction, mental health, and academic performance.

Platform growth rose sharply after 2010.

Male and female user-count growth shows similar upward patterns across Twitter and Facebook.

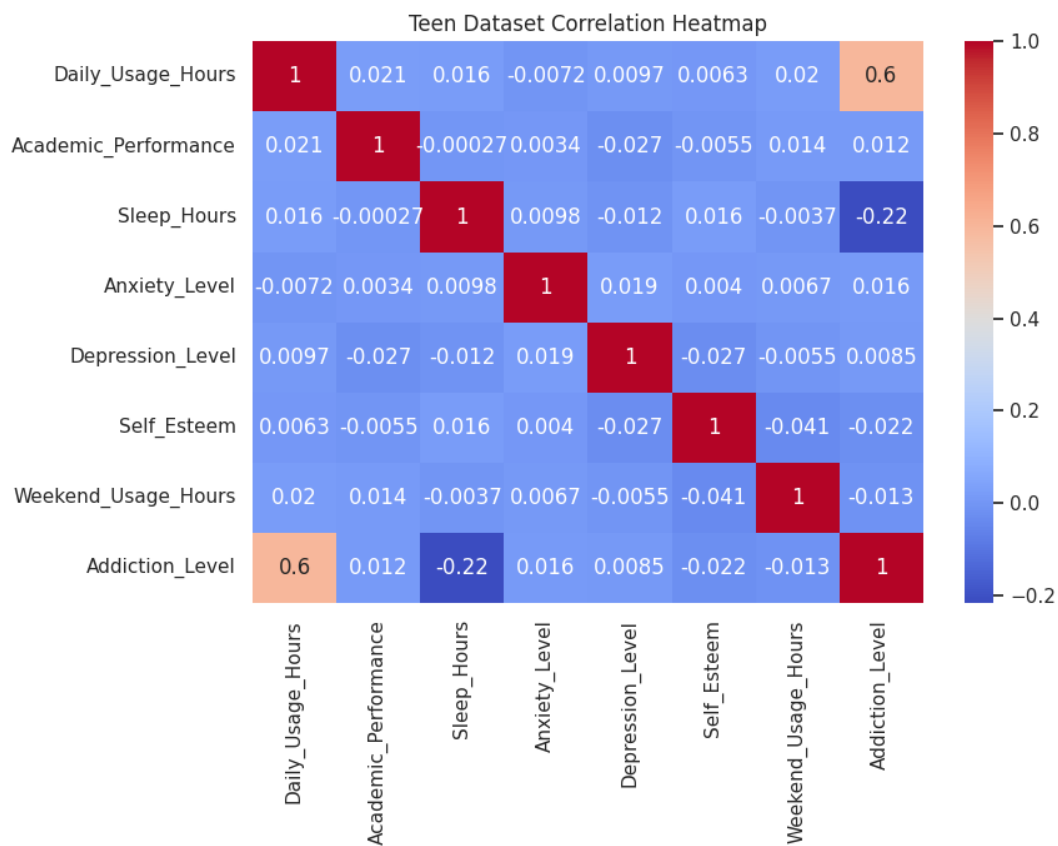


Why this matters

The national data shows major social-media adoption growth, but it cannot explain individual outcomes by itself. That is why the teen and student datasets are necessary.

For teens, usage connects most clearly to addiction.

Daily usage hours showed the strongest relationship with addiction level; academic performance was less directly connected.



Teen dataset correlation heatmap

0.60

Daily usage hours ↔ addiction level

-0.22

Sleep hours ↔ addiction level

Main takeaway: addiction and emotional wellbeing are more closely linked to phone usage than academic outcomes in the teen dataset.

Anxiety and depression move together.
More addicted teens tend to sleep less.
Grades may depend on other factors beyond screen time.

Screen time alone was not the whole story.

The teen data suggests social-media impact is more complex than “more hours = worse grades.”

72–75

average academic score range across
usage groups

0.60

usage ↔ addiction correlation

moderate

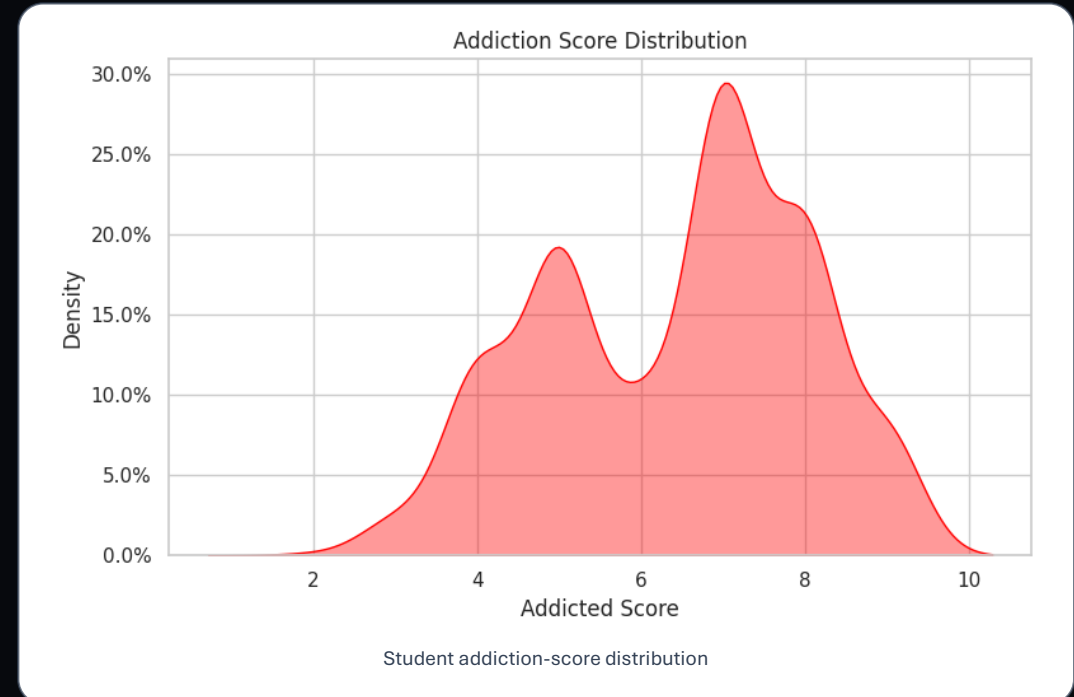
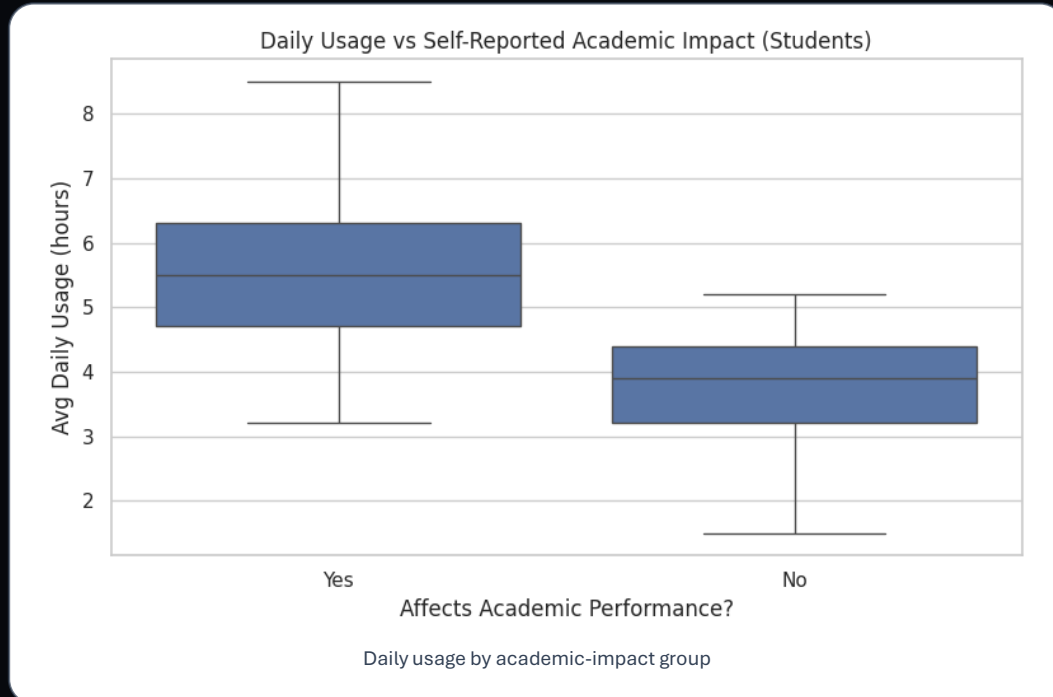
anxiety and depression moved
together

Interpretation

For teens, the more important story is not just time spent online. Addiction, sleep, and emotional wellbeing appear more meaningful than daily usage alone.

Among students, the academic impact was clearer.

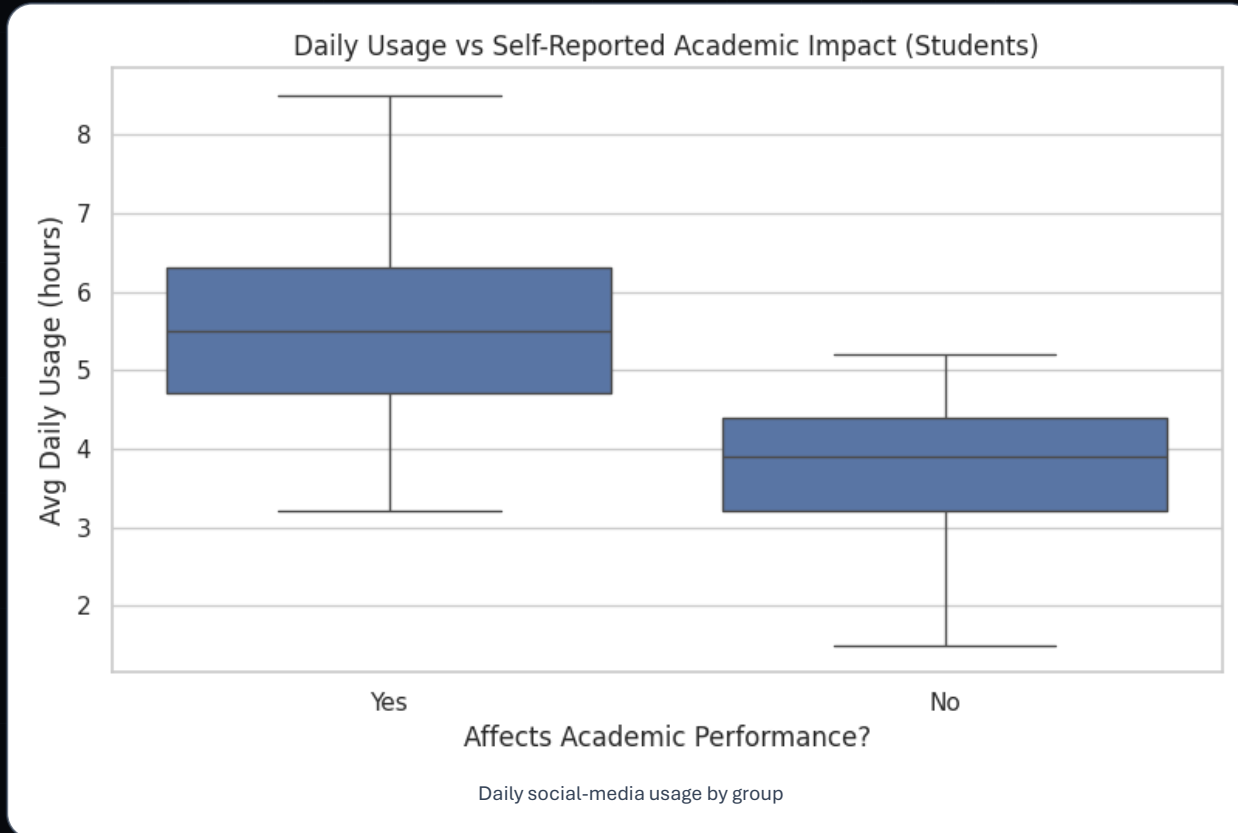
Students who reported academic impact tended to use social media more and show higher addiction scores.



Students in the “Yes, it affects academics” group had a higher median daily usage and wider spread, including heavy-use outliers.

Heavier daily use aligned with reported academic impact.

The boxplot shows a clear separation between students who report academic impact and those who do not.



~5.5 hrs

median usage for students reporting
academic impact

~4 hrs

median usage for students
reporting no impact

The “Yes” group uses social media more daily.
The “Yes” group has wider usage variation.
Some students report 7–8 hours per day.

Addiction intensity was a major signal.

Moderate to high addiction scores were common, and higher addiction was strongly tied to poorer outcomes in the student data.



7.4

average addiction score for students
reporting academic impact

4.5

average addiction score for students
reporting no impact

-0.95

addiction score ↔ mental health score

The strongest student finding: addiction level appears more explanatory than daily use alone.

Teens and students showed different patterns.

The analysis suggests that age/context changes how social-media behavior appears in the data.

Teens

Academic scores stayed in a narrow range.
Usage linked to addiction and emotional wellbeing.
Screen time alone did not strongly explain grades.

Students

Academic impact split into clearer groups.
Higher addiction was linked to worse outcomes.
Heavy use tied to sleep and mental health patterns.

A simple model confirmed the EDA pattern.

Logistic regression predicted academic impact from behavioral and wellbeing variables.



100%

reported model accuracy

Note: 100% accuracy is very high, so it should be explained carefully as a sign of strong patterns in this dataset, not automatic real-world perfection.

Final takeaways.

The strongest findings were about addiction, academic impact, sleep, and mental health — not just raw screen time.

Heavy use

Students with heavier daily use were more likely to report academic impact.

Teens

Teen grades did not shift much by usage group, suggesting more complex causes.

Addiction

Higher addiction scores were tied to poorer academic and mental-health outcomes.

Awareness

Moderation and awareness are important, especially for high-addiction patterns.

Ready for the portfolio.

A cleaner project story that can be shown on a personal website, résumé, or internship application.

Project card copy

Social Media Impact Study

Analyzed social media usage, mental health, addiction, and academic performance using EDA, heatmaps, boxplots, and logistic regression.

Python

Pandas

EDA

Visualization

Logistic Regression

What this project shows

Data cleaning and exploratory analysis
Correlation interpretation and careful wording
Clear visualization of student behavior patterns
Basic machine-learning classification workflow